

- **Phenol-formaldehyde Resins (Bakelite)** These are made by the reaction of phenol and formaldehyde in the basic medium. Bakelite is a cross-linked thermosetting polymer. It is used for making combs, fountain pen barrels, electrical goods making electrical appliances, handles of crockery, etc. Sulphonated bakelites are used as ion-exchange resins for softening of hard water.
- **Rubber** It is a polymer of isoprene (2-methyl buta-1, 3-diene). It is insoluble in water, dilute acids and alkalis, absorbs a large amount of water, has low tensile strength and elasticity.
- **Neoprene** It is a synthetic rubber which resembles natural rubber in its properties. It is obtained by the polymerisation of chloroprene (2-chlorobuta-1, 3-diene). It is superior to natural rubber in its stability. Easy availability of raw materials from products of petroleum refining industries lead to growth of synthetic rubber. It is generally used for making hoses, shoe heels, stoppers, etc.
- **Buna-S** It is a copolymer of 1, 3-butadiene and styrene. It is also known as SBR (Styrene-Butadiene Rubber). It has slightly less tensile strength than natural rubber. It is used in the manufacturing of automobile tyres, rubber soles, etc.
- **Terylene or dacron or phthalate** It is a polymer of ethylene glycol and terephthalic acid, i.e. it is a polyester. It is used for making wash and wear fabrics, tyre cords, safety belts, tents, etc.
- **Vulcanisation of Rubber** It is the process of heating of rubber with sulphur. It makes the rubber hard, strong and more elastic.
- **Kevlar** It is a polymer of terephthalic acid and 1,4-diaminobenzene so, it is a polyamide. It is used for making bulletproof vests.
- **Lexan or polycarbonate** It is a polymer of diethyl carbonate and bi-phenol. It is used in making bulletproof windows and safety helmets.
- **Polyurethanes** It is a polymer of toluene diisocyanate and ethylene glycol. It is used for making washable and long lasting mattresses, cushions, pillows, etc.
- **Lucite** It is a polymer of methyl methacrylate. It is used for making contact lenses. Plexiglass, acrylite or perspex are other terms used for lucite.
- **Orlon or Acrilon Polyvinyl Cyanide** It is a polymer of vinyl cyanide or acrylonitrile. It is used as a substitute for wool in making synthetic blankets (acrylic fibres).

FERTILISERS

Repeated cultivation of crops makes the soil deficient in some elements, mainly nitrogen, phosphorus and potassium. The substances which are added to the soil to make up the loss of these elements are called **fertilisers**.

Some Important Fertilisers

- **Basic Calcium Nitrate** (Nitrate of lime) $[CaO \cdot Ca(NO_3)_2]$ It is a good nitrogeous fertiliser, used in acidic soil.
- **Ammonium Sulphate** $[(NH_4)_2SO_4]$ It contains 21.2% nitrogen. Its repeated use makes the soil acidic and unfit for the germination of seeds. Hence, addition of lime to the soil becomes necessary.
- **Calcium Cyanamide** $[CaCN_2]$ (Nitrolim) It is black in colour because of the presence of carbon. It contains 19% nitrogen.
- **Urea** $[NH_2CONH_2]$ (Carbamide) It contains 46.6% nitrogen. It is the best nitrogeous fertiliser as it leaves only carbon dioxide after ammonia has been assimilated by plants. Its repeated use does not change the pH of the soil.
- **Superphosphate of Lime** $[Ca(H_2PO_4)_2 + 2CaSO_4 \cdot H_2O]$ It is phosphatic fertiliser.
- **Vermicompost** It is the product or process of composting by using various species of worms, usually red wigglers, white worms and earthworms to create a heterogeneous mixture of decomposing vegetables or food, bedding material and vermicast containing water soluble nutrients. Vermicompost is an excellent, nutrient rich organic fertiliser and soil conditioner.
- Magnesium, manganese and sulphur are the macronutrients which are provided by the inorganic fertilisers.

FUELS

The substances which on combustion produces energy in the form of heat, is called a fuel, e.g. coal, wood, kerosene, petrol, diesel, cooking gas, etc. Fuel efficiency is expressed in terms of calorific value (kJg^{-1}).

- Sulphur compound (ethyl mercaptan or thioethanol) is added to odourless LPG (Liquefied Petroleum Gas) for imparting a detectable smell to the gas.
- LPG contains mainly butane alongwith some propane. CNG (Compressed Natural Gas) contains mainly methane and is used as a fuel in many vehicles because it produces less pollutants therefore, it is a better fuel than petrol or diesel.
- Petrol is used as a fuel to run cars and aeroplanes whereas diesel is used to run trucks, buses, trains and ships.
- Quality of petrol or gasoline is expressed in terms of **octane number** and of diesel in terms of **cetane number**.
- Synthetic rubber, liquid ammonia, liquid hydrogen are used as propellants.
- Acetylene is a fuel used in gas welding. It is used to weld and cut the metals that have a temperature of 300°C of a flame.

Renewable and Non-renewable Resources

- Natural resources such as coal, petroleum and natural gas take thousands of years to form naturally and cannot be replaced as fast as they are being consumed. It is a non-renewable resource.
- Natural resources such as solar energy, water and fisheries are renewable resources.

Petroleum

It is dark oily liquid also called rock oil, mineral oil, crude oil or black gold. On fractional distillation, it gives different substances at different temperatures.

Biogas or Gobar Gas

The gaseous mixture obtained by the degradation of animal and plant wastes by anaerobic microorganisms in the presence of water is called biogas. It is a convenient fuel for domestic use. It is used for street lighting.

- Constituents of biogas are methane (45-70%), carbon dioxide, hydrogen, hydrogen sulphide.
- Gobar gas is produced by the fermentation of cow dung. It contains CH_4 (main component), CO_2 and H_2 .
- **Note** Bagasse is the fibrous matter remains after sugarcane stalks are crushed to extract their juice. It is used as a biofuel and as a substitute for wood in many tropical and subtropical countries for the production of pulp, paper, board, etc.

Flame

A flame is a region where combustion of gaseous substances take place. Innermost region of flame is

black because of the presence of unburned carbon particles. Middle region is yellow luminous due to partial combustion. Outermost region is blue due to complete combustion of fuel which is the hottest part of the flame.

Paint

It is a mixture of four ingredients, *viz*, binder, solvent, pigment and additives (like driers, plasticisers, emulsifiers, corrosion inhibitors, etc).

- Binders are polymers (resins) forming a continuous film on the surface of the substrate. These are responsible for good adhesion. Resins, chlorinated rubber, latex (PVA) are the examples of binders.
- Solvent (water or organic solvents like toluene, ketone, alcohol, etc.) is a medium where all other ingredients of paint are dispersed in molecular form (true solutions) or as colloidal dispersion (emulsions).
- Pigments are responsible for imparting colour to the paint. It may also protect the substrate from UV light.
- Ultramarine blue is the natural pigment made up of semiprecious mineral lapis lazuli. It is resistant towards fading.

Pigment added	Colour of paint
Chromium oxide	Green
Vermillion or cuprous oxide	Red
Cobalt oxide	Blue
Titanium dioxide	White
Benzidine yellow	Yellow to red
Iron (II) oxide	Black

ENVIRONMENT AND ITS POLLUTION

Air

- Composition of air is nitrogen (78.084%), oxygen (20.946%), argon (0.934%), carbon dioxide (0.033%) and small amount of other gases. Region of air present around the earth is called atmosphere.
- Main layers from the surface of earth towards upward

(a) Troposphere	(b) Stratosphere
(c) Mesosphere	(d) Thermosphere
- Most of the atmospheric air is present in troposphere.
- Ozone layer present in the stratospheric region (at a height of about 20 to 40 km) protects the living beings from the harmful UV radiations coming from the sun. Due to high concentration of ozone, this region is also known as ozonosphere.
- In 1775, French scientist Lavoisier performed experiments on composition of air.

Air Pollution

It is due to the presence of foreign undesirable substances in air. The substance which causes pollution is known as pollutant. Main air pollutants are SO_2 , CO, nitrogen oxides, particulates, etc.

Carbon monoxide CO combines with the haemoglobin of the blood about 300 times more readily as compared to oxygen and forms a stable carboxyhaemoglobin complex due to which supply of oxygen is restricted which results in headache, nervousness, asphyxia and even death in some cases.

Major gaseous air pollutants:

- Oxides of carbon
- Oxides of nitrogen
- Oxides of sulphur
- Hydrocarbon (esp CH_4)

Methyl isocyanate (CH_3NC) is responsible for the Bhopal gas tragedy (in 1984). In which thousands of people were killed. It was a case of serious air pollution in which MIC (Methyl Isocyanate) gas released from a pesticide manufacturing plant of union carbide.

Smog

The word smog is derived from smoke and fog. It is of two types :

1. **Classical Smog** It is also called London smog. It is formed in cool humid climate and is reducing in nature.
2. **Photochemical Smog** It is also called Los Angeles smog. It is formed in warm, dry and sunny climate and is oxidising in nature.

Particulates

These are minute solid particles and liquid droplets dispersed in air such as mists, dusts, smoke, etc.

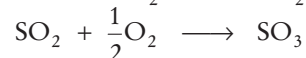
Diseases Caused by Particulates

Disease	Cause
Pneumoconiosis	Coal dust
Silicosis	Silica (from ceramics, glass and pottery industry)
Black lung disease	Coal mines
White lung disease	Textile industries
Asbestosis	Asbestos
Byssinosis	Cotton fibre dust
Melanosis	Hyperpigmentation due to increased concentration of melanin

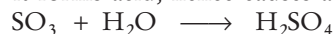
Consequences of Air Pollution

- **Greenhouse effect** It is the heating up of earth and its objects due to trapping of infrared radiations (IR) by greenhouse gases present in atmosphere like CO_2 , CH_4 , NO , O_3 , chlorofluorocarbon and water vapours.
- **Global Warming** It is due to increased concentration of greenhouse gases. It may lead to melting of ice caps and glaciers, spreading of several infectious diseases like malaria, sleeping sickness, etc.
- Since, the late 19th century, the temperature of atmosphere increases due to the increase in CO_2 concentration which is responsible for the expansion of water in the ocean.
- **Acid Rain** The pH of normal rain water is 5.6-6.5 due to the dissolution of carbon dioxide from the atmosphere. When the pH of rain water is below 5, it is called acid rain (by Robert August).
- The main cause of acid rain is oxides of sulphur and nitrogen (H_2SO_4 and HNO_3).

- Burning of sulphur alongwith coal results in the formation of oxides of sulphur as

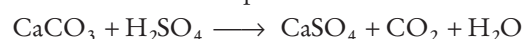


when sulphur trioxide (SO_3) reacts with water of clouds it forms acid, hence causes acid rain



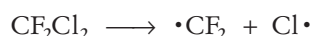
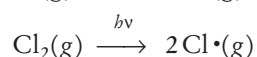
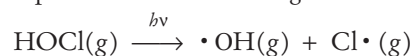
- Acid rain damages the marble buildings (Taj Mahal) and monuments, corrodes metal pipes and results in several diseases.

Acid rain reacts with buildings made from limestone and causes its decomposition

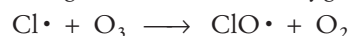


- **Ozone Layer Depletion** Thinning of ozone layer because of its reaction with chlorine free radicals usually generated by chlorofluorocarbons is called **ozone layer depletion**.

- Hypochlorous acid and chlorine on photochemical decomposition also produce nascent chlorine ($\text{Cl}\cdot$ or chlorine free radical) which causes ozone depletion in the following manner:



- The released chlorine free radical reacts with ozone breaking it into molecular oxygen.



Water Pollution

Water pollution is due to the presence of foreign substances, i.e. water pollutants in water.

- Cholera, dysentery, typhoid, etc., are water-borne diseases caused by bacteria.
- Mercury causes minamata disease, chromium and arsenic causes cancer and cadmium causes *itai-itai* disease.
- Presence of excessive fluoride leads to dental fluorosis, i.e. developmental disturbance of dental enamel.
- The usual effect of agricultural run off due to the excessive increase of nutrients such as nitrogen, phosphorus etc., in an ecosystem like ponds or results in excessive algal growth in affected water bodies. It is called **eutrophication**.
- For clean water, DO (Dissolved Oxygen) is 5-6 ppm and BOD (Biochemical Oxygen Demand) is less than 5 ppm.
- Recycling, sewage treatment, incineration are some strategy to control water pollution.
- **Presbycusis** is gradual hearing loss in both ears that commonly occurs in aged people but noise pollution also lead to it.

> PRACTICE EXERCISE

Matter

1. During evaporation, particles of a liquid change into vapours only
(a) from the surface (b) from the bulk
(c) from both surface and bulk
(d) None of the above

2. Arrange the following in increasing order of density.

1. Air 2. Honey
3. Water 4. Iron
5. Chalk

Codes

- (a) $1 < 2 < 3 < 4 < 5$
(b) $1 < 2 < 3 < 5 < 4$
(c) $1 < 3 < 2 < 5 < 4$
(d) $1 < 3 < 2 < 4 < 5$

3. Which one of the following mixtures is homogeneous?

- (a) Starch and sugar
(b) Methanol and water
(c) Graphite and charcoal
(d) Calcium carbonate and calcium bicarbonate

4. Which one of the following properties changes with valency?

- (a) Atomic weight (b) Equivalent weight
(c) Molecular weight (d) Density

5. 44 g of carbon dioxide at STP contains

- (a) 6.02×10^{23} molecules
(b) 3.01×10^{23} molecules
(c) 6.02×10^{24} molecules
(d) 3.01×10^{24} molecules

6. Which of the following is a physical change?

- (a) Decomposition (b) Oxidation
(c) Sublimation (d) Reduction

7. **Statement I** Ice floats over water.
Statement II Density of liquids is generally lower than that of the solids.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

8. Which of the following is a chemical change?

1. Burning of coal in air
2. Crystallisation of table salt from sea water.

Select the correct answer using the codes given below.

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

9. Which of the following is not a mixture?

1. Toothpaste 2. Baking soda
3. Vinegar

Choose the correct option

- (a) Only 1 (b) Only 2
(c) 1 and 3 (d) 2 and 3

10. On addition of salt to water, its

1. boiling point increases
2. boiling point decreases
3. boiling point is not affected
4. freezing point decreases

The correct option(s) is/are

- (a) 1 and 4 (b) 2 and 4
(c) Only 2 (d) Only 3

11. Match the following Columns.

Column I (Process)	Column II (Changes)
A. Evaporation	1. Liquid into gas
B. Sublimation	2. Gas into liquid
C. Freezing	3. Solid into gas
D. Melting	4. Solid into liquid
	5. Liquid into solid

Codes

- A B C D
(a) 2 1 3 4
(b) 1 3 5 4
(c) 4 3 2 1
(d) 5 3 1 2

> Previous Years' Questions

12. Iodised salt is a ☑ 2012 II

- (a) mixture of potassium iodide and common salt
(b) mixture of molecular iodine and common salt
(c) compound formed by the combination of potassium iodide and common salt
(d) compound formed by molecular iodine and common salt

13. **Statement I** All compounds contain more than one element.

Statement II All compounds are heterogeneous mixtures.

☑ 2012 II

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
(b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

14. A liquid initially contracts when cooled down to 4°C but on further cooling down to 0°C , it expands.

The liquid is ☑ 2013 II

- (a) alcohol (b) water
(c) molten iron (d) mercury

15. A metal screw top on a glass bottle which appears to be stuck could be opened by using the fact that ☑ 2014 I

- (a) the metal expands more than the glass when both are heated
(b) the metal and glass expand identically when heated
(c) the metal shrinks when heated
(d) both metal and glass shrink when cooled

16. Which one among the following is an example of chemical change? ☑ 2014 II

- (a) The melting of an ice cube
(b) The boiling of gasoline
(c) The frying of an egg
(d) Attraction of an iron nail to a magnet

17. Which one among the following compounds has same equivalent weight and molecular weight?

☑ 2015 I

- (a) H_2SO_4 (b) CaCl_2
(c) Na_2SO_4 (d) NaCl

18. Creation of something from nothing is against the law of ☑ 2015 I

- (a) constant proportions
(b) conservation of mass energy
(c) multiple proportions
(d) conservation of momentum

19. Match the following Columns.

☞ 2015 II

Column I (Exponent)	Column II (Law)
A. John Dalton	1. Law of definite proportion by volume
B. Joseph Proust	2. Law of multiple proportion
C. Antoine Lavoisier	3. Law of definite proportion by weight
D. Joseph Louis Gay Lussac	4. Law of conservation of mass

Codes

A B C D	A B C D
(a) 2 3 4 1	(b) 2 4 3 1
(c) 1 4 3 2	(d) 1 3 4 2

20. Which one of the following is an example of chemical change?

☞ 2016 I

- (a) Burning of paper
- (b) Magnetisation of soft iron
- (c) Dissolution of cane sugar in water
- (d) Preparation of ice cubes from water

21. A piece of ice, 100 g in mass is kept at 0°C. The amount of heat it requires to melt at 0°C is (take latent heat of melting of ice to be 333.6 J/g).

☞ 2016 I

- (a) 750.6 J
- (b) 83.4 J
- (c) 33360 J
- (d) 3.336 J

22. After a hot sunny day, people sprinkle water on the roof-top because

☞ 2016 I

- (a) water helps air around the roof-top to absorb the heat instantly
- (b) water has lower specific heat capacity
- (c) water is easily available
- (d) water has large latent heat of vaporisation

Atomic Structure

23. Neutrons are present in all atoms except

- (a) He
- (b) C
- (c) H
- (d) Ne

24. Positive rays in a discharge tube with perforated cathode travel

- (a) parallel to the anode
- (b) parallel to the cathode
- (c) from cathode to anode
- (d) from anode to cathode

25. The increasing order (lowest first) for the value of e/m (charge/mass) is

- (a) e, p, n, α
- (b) n, p, e, α
- (c) n, p, α, e
- (d) n, α, p, e

26. Which of the following properties of the element can be a whole number?

- (a) Atomic mass
- (b) Atomic number
- (c) Atomic radius
- (d) Atomic volume

27. There are 15 protons and 16 neutrons in the nucleus of an element. The element is represented by the symbol

- (a) $^{16}_{15}X$
- (b) $^{30}_{15}X$
- (c) $^{31}_{16}X$
- (d) $^{31}_{15}X$

28. Isotopes of the same elements have

- (a) same number of neutrons
- (b) same number of protons
- (c) same atomic mass
- (d) different chemical properties

29. The atomic number of an element of mass number 27, which has 13 neutrons is

- (a) 10
- (b) 14
- (c) 12
- (d) 13

30. Which of the following statement(s) is/are true?

- (a) Atomic number is always greater than mass number
- (b) Mass number may be equal to atomic number
- (c) Mass number is equal to the sum of number of protons and electrons
- (d) Mass number never be equal to atomic number

31. How many electrons are present in the M shell of an atom of an element with atomic number 24?

- (a) 5
- (b) 6
- (c) 12
- (d) 14

32. The atom of an element contains 2 electrons in its M shell. The element is

- (a) aluminium
- (b) sodium
- (c) chlorine
- (d) magnesium

33. While performing cathode ray experiment, it was observed that there was no passage of electric current under normal conditions. Which of the following can account for this observation?

- (a) Dust particles are present in air
- (b) Carbon dioxide is present in air
- (c) Air is a poor conductor of electricity under normal conditions
- (d) All of the above

34. If K and L shells of an atom are full, then what would be the total number of electrons in the atom?

- (a) 8
- (b) 10
- (c) 15
- (d) 6

35. Taking into account three isotopes of hydrogen and three isotopes of oxygen occurring in nature, how many different kinds of water molecules can we expect on the earth?

- (a) Six
- (b) Nine
- (c) Twelve
- (d) Eighteen

36. Composition of two nuclei is shown below.

	X	Y
Atomic number	7	9
Mass number	15	17

X and Y are

- (a) isobars
- (b) isotones
- (c) isotopes
- (d) isomers

37. Statement I X-rays are not deflected by electric field.

Statement II X-rays travels with the velocity of light.

Codes

- (a) Both the Statements are correct and Statement II is the correct explanation of Statement I
- (b) Both the Statements are correct but Statement II is not the correct explanation of Statement I
- (c) Statement I is correct but Statement II is incorrect
- (d) Statement I is incorrect but Statement II is correct

38. Consider the following statements

1. An atom consists of a positively charged sphere and the electrons are embedded in it.
2. The negative and positive charges are equal in magnitude, therefore the atom as a whole is electrically neutral.

Select the correct statement(s).

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) None of these

39. Consider the following statements.

1. Cathode rays are originated from cathode and anode rays from anode.
2. Both travel in straight line.
3. Electrons are the fundamental particle of an atom.
4. e/m ratio of anode rays does not depend upon the nature of gas taken in discharge tube.

Select the correct statement(s).

- (a) 1 and 4
- (b) 2 and 3
- (c) 1, 3 and 4
- (d) 2, 3 and 4

40. Match the following Columns.

Column I	Column II
A. Proton	1. Chadwick
B. Neutron	2. Millikan
C. Charge of electron	3. Goldstein
D. Shelled nature of atom	4. Rutherford
	5. Madam Curie